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A Safety-Gated, Hindi-First Agent for Appointments and Urgency Triage

Mid-Submission Report

LMA Group Project 2026

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Abstract

Swasthya Saathi is a research prototype of a clinic front-desk agent for patients who write in Devanagari Hindi, romanised Hindi (Hinglish) or English. Its central idea is to take two decisions out of the language model: a small fine-tuned *safety gate* decides urgency, and a deterministic *action guard* approves state-changing tool calls. This report covers the triage half of the project: a script-parallel triage dataset, four triage systems, and a first set of ablations. A QLoRA-tuned gate reaches emergency recall 0.913 with over-triage 0.067 on the held-out test split, and at a matched over-triage budget it finds substantially more emergencies than a prompted gate. The clinic simulator, action guard and the 2×2 agent ablation are not yet built.

1 Status summary

The triage half of Tier 1 is finished and evaluated on the held-out test split; the clinic simulator, action guard and 2×2 agent ablation have not been started.

Area	Status
D3 script-parallel triage set: 150 evaluation cases and 300 training cases, each in English, Devanagari and Hinglish	Done
Parity check between the three renderings; train/test overlap check	Done
Triage systems: keyword baseline (B0), MuRIL (B1), prompted Qwen2.5-3B (V0), QLoRA gate (G1)	Done, 3 seeds for trained models
H1 comparison at a fixed over-triage rate; training-script ablation	Done
Clinic simulator D4, four tools, action guard, 2×2 system arms	Not started
Clinician review of the rubric and emergency templates	Not started

Headline results (test split, 42 emergency cases per script):

- The QLoRA gate G1, trained on all scripts, reaches emergency recall 0.913 ± 0.035 with over-triage 0.067 ± 0.011 and macro-F1 0.812 ± 0.037 (mean $\pm$ sd over 3 seeds).
- At a matched over-triage budget of 10% set on the dev split, G1 finds 0.820 ± 0.030 of emergencies, against 0.627 for the 8-shot prompted gate and 0.508 for the zero-shot one. Paired confidence intervals exclude zero for all three G1 seeds, so H1 is supported on this test set.
- Script gap: zero-shot Qwen catches 0.905 of English emergencies but only 0.571 of Hinglish ones. G1 narrows this to 0.952 / 0.929 / 0.857 (English / Devanagari / Hinglish).
- Training on English only does not transfer: G1 trained on English alone drops to 0.595 emergency recall, with Devanagari 0.524 and Hinglish 0.405.

2 Progress to date

The triage lane has reached the proposal's week-4 milestone ("train first gates, baseline"), while the agent lane (D4 simulator, tools, guard, harness) is behind and is the main remaining work.

- **Pipeline.** A Python repository with a shared library (labels, metrics, evaluation harness, prompts), validators for the datasets, standalone GPU scripts for MuRIL, the prompted Qwen gate and the QLoRA gate, and scoring scripts that turn raw predictions into result tables and figures.
- **Workflow.** Code and analysis run locally; all GPU work runs on Kaggle (T4 GPUs). Code and data are bundled and uploaded as one Kaggle dataset, outputs are downloaded and scored locally, and every run stores its configuration next to its predictions.
- **Reproducibility.** Fixed seeds (13, 42, 2026), per-run config files, and committed result files for every reported number. Discarded or superseded runs are kept in an archive folder rather than deleted.
- **Evaluation protocol.** Confidence intervals are bootstrapped over cases, not renderings, because the three script versions of one case are not independent. Primary metric: emergency recall, always reported together with over-triage and macro-F1.

3 Datasets

All data are synthetic. D3 has 450 cases (150 for evaluation, 300 for training), each written in English, Devanagari Hindi and Hinglish; D4 (the clinic simulator) does not exist yet.

Dataset	Size	Labels	Notes
D3 evaluation	150 cases × 3 scripts = 450 renderings	EMERGENCY 60, URGENT 36, ROUTINE 30, INSUFFICIENT 24	21–22 cases per category; 24 minimal pairs; split by case into dev (45 cases, 18 emergencies) and test (105 cases, 42 emergencies), stratified, keeping pairs together
D3 training	300 cases × 3 scripts = 900 renderings	EMERGENCY 120, URGENT 75, ROUTINE 60, INSUFFICIENT 45	42–43 cases per category; no minimal pairs; written by a different team member than the evaluation set
D4 clinic simulator	planned: about 60–70 scenarios	programmatic goal state	not started

How D3 was built. Each case starts from a list of clinical cues (symptoms, severity, duration, age group, who is writing). The writer then composed the three renderings so that each contains exactly those cues; spelling, politeness and filler words may differ. This differs from the proposal, which planned to render all three forms programmatically from one structured specification: here a person wrote every rendering, and the cue list is the checklist.

Parity check. A second team member listed, for each rendering read on its own, which cues it contains and whether it adds any unlisted clinical detail. On the first pass 143 of 150 rows (95.3%) matched the cue list. All 7 mismatches were small unlisted details (intensifiers such as "very", a quantity phrase, a counselling request, an implied young age). They were adjudicated

by expanding the cue list (3 rows) or editing the text (5 rows; one row had both), after which all rows match by construction. The original file is archived, so the 95.3% first-pass figure is the honest one to quote. This is a cue-parity check, not an independent agreement measure on the urgency labels, which has not been run.

Overlap between training and evaluation text. Using character 5-gram Jaccard similarity between each training rendering and every evaluation rendering of the same script, the mean over training cases was 0.156 and the maximum 0.415; no case reached 0.5, none was dropped, and none of the 300 training cases shares an exact cue set with an evaluation case.

Known weaknesses. The urgency rubric follows public triage guidance but has not been reviewed by a clinician; the training text contains some typos and missing spaces; all data were written by the project team.

4 Experimental setup

Four triage systems are compared on the same four-way task (EMERGENCY, URGENT, ROUTINE, INSUFFICIENT); only the QLoRA gate G1 is the proposed method, the others are baselines.

ID	System	Training and settings
B0	Keyword lexicon (English, Devanagari, Hinglish terms)	None. Written before seeing the data and frozen; no negation handling
B1	MuRIL (`google/muril-base-cased`) with a classification head	D3 train, 10 epochs, learning rate 5×10^{-5}, batch size 8; seeds 13, 42, 2026
V0	Qwen2.5-3B-Instruct, 4-bit, prompted	No training; zero-shot and 8-shot (2 examples per label taken from D3 train)
G1	Qwen2.5-3B-Instruct + QLoRA (rank 16, alpha 32)	D3 train, 3 epochs, learning rate 2×10^{-4}, batch 4 with 4 accumulation steps; seeds 13, 42, 2026

V0 and G1 share one read-out: the model is asked to answer with a single letter (A to D), and the logits of those four letters at the answer position are turned into probabilities, so there is no free-text parsing and both give probabilities that the threshold analysis below can use (they are not yet calibrated).

Metrics. Emergency recall (primary), over-triage (non-emergencies predicted EMERGENCY), under-triage (urgent or emergency cases predicted less severe), macro-F1, and the cross-script gap (highest minus lowest emergency recall across the three scripts). 95% confidence intervals use 1,000 bootstrap resamples over cases. Minimal-pair flip accuracy and calibration error are computed by the harness but not yet reported.

What was tuned, and on what. No setting was changed after test results were first seen; the test split was scored again only to add the later QLoRA seeds and ablation runs.

- MuRIL received one round of tuning on dev only. The first configuration (5 epochs, learning rate 3×10^{-5}, batch 16) underfit; one seed was tried at 15 epochs, and 10 epochs was chosen because dev emergency recall fell after epoch 5.

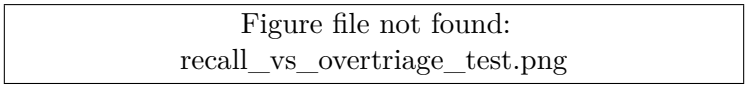


Figure 1: Emergency recall against over-triage, one point per run (test split). Points toward the top left are better; a high recall reached through a high over-triage rate is not.

- G1 was not tuned: the first configuration is the reported one. If anything, the baseline therefore received more tuning than the proposed method.
- Early stopping on emergency recall and calibration on dev, both planned in the proposal, are not applied yet.

5 Baseline results

On the test split G1 has the best macro-F1 and the lowest over-triage of all systems while keeping emergency recall above 0.9, and it is far more stable across seeds than MuRIL.

System	Runs	Emergency recall	Over-triage	Under-triage	Macro-F1	Scrip
B0 keyword	1	0.484	0.143	0.415	0.450	0.1
V0 Qwen, zero-shot	1	0.738	0.132	0.236	0.547	0.3
V0 Qwen, 8-shot	1	0.802	0.127	0.231	0.691	0.1
B1 MuRIL, all scripts	3	0.884 ± 0.119	0.289 ± 0.307	0.077 ± 0.080	0.630 ± 0.120	$0.040 \pm$
G1 QLoRA gate, all scripts	3	$\mathbf{0.913 \pm 0.035}$	$\mathbf{0.067 \pm 0.011}$	0.065 ± 0.029	$\mathbf{0.812 \pm 0.037}$	$0.095 \pm$

Table 1: Test split, 126 emergency renderings (42 cases per script). Mean ± sd over seeds; the prompted systems are deterministic, so one run each. Decisions use the model's most likely class; the matched comparison in the next section removes the effect of different operating points.

Three things to read carefully:

- **Recall alone misleads.** One MuRIL seed (2026) has recall 1.0 only because it labels about 64% of non-emergencies as EMERGENCY (macro-F1 0.49); this is why over-triage and macro-F1 are reported next to recall.
- **MuRIL is unstable across seeds.** The same configuration gives macro-F1 0.70, 0.70 and 0.49; the third run barely trained (final training loss 0.51 against about 0.02 for the others).
- **Same seed, different outcome.** Seed 13 of G1 was trained twice on Kaggle. The first run scored recall 0.944, over-triage 0.116, macro-F1 0.810; the second, reported above, scored 0.937, 0.064, 0.850. Training a 4-bit model on a GPU is not bit-exact, so differences of a few points between single runs should not be over-read. The first run's results are archived.

6 H1: tuned gate versus prompted gate at a fixed over-triage rate

H1 is supported on this test set: at the same over-triage budget, the QLoRA gate finds 0.19 more of the emergencies than the 8-shot prompted gate (0.820 against 0.627) and 0.31 more than the zero-shot one (0.508).

Method. For each system we choose, on the dev split only, the smallest threshold on $P(EMERGENCY)$ whose dev over-triage is at most the budget. On the test split we then

predict EMERGENCY whenever $P(EMERGENCY)$ reaches that threshold and measure emergency recall. Budgets of 5%, 10% and 20% are shown; 10% is the primary one.

System	Runs	Budget 5%	Budget 10%	Budget 20%
MuRIL, all scripts	3	0.278 ± 0.290	0.476 ± 0.414	0.550 ± 0.443
V0 Qwen, zero-shot	1	0.421	0.508	0.754
V0 Qwen, 8-shot	1	0.508	0.627	0.754
G1 QLoRA gate	3	$\mathbf{0.704 \pm 0.097}$	$\mathbf{0.820 \pm 0.030}$	$\mathbf{0.926 \pm 0.046}$

Table 2: Test emergency recall at the threshold chosen on dev (mean ± sd over seeds).

Paired comparison at the 10% budget. The difference in recall between each G1 seed and each prompted gate, with a 95% interval from a bootstrap over test cases (the same resampled cases for both systems):

- G1 minus 8-shot: +0.206 [+0.100, +0.314] (seed 13), +0.159 [+0.048, +0.272] (seed 2026), +0.214 [+0.120, +0.316] (seed 42).
- G1 minus zero-shot: +0.325 [+0.232, +0.415] (seed 13), +0.278 [+0.187, +0.362] (seed 2026), +0.333 [+0.217, +0.438] (seed 42).

All six intervals exclude zero.

Caveats.

- The 10% budget was chosen after earlier results were seen, so it is not a pre-registered threshold; the 5% and 20% budgets point the same way.
- Dev has only 81 non-emergency renderings, so a 5% budget means at most 4 false alarms and the threshold is coarse.
- Because the threshold comes from dev, the realised test over-triage at the 10% budget was lower (0.026 to 0.042 across systems), and at 20% it differed more between systems (0.093 for G1, 0.101 for 8-shot, 0.164 for zero-shot), which favours the zero-shot gate at that budget.
- The intervals reflect which test cases were sampled, not variation between training seeds, which is shown separately by the ± values.
- MuRIL scores poorly and very unevenly here, probably because its probabilities are extreme (not yet verified), so a threshold cannot trade recall for false alarms; this supports the proposal's plan to calibrate on dev.
- The proposal planned McNemar tests; we used a paired bootstrap over cases because the three renderings of a case are not independent.

7 Ablations

Training the QLoRA gate on English alone does not carry over to Devanagari or Hinglish, while training on Hinglish alone transfers to all three scripts almost as well as training on all of them.

Training-script ablation. Each model was trained on one script only (300 renderings) or on all three (900 renderings), and tested on all three scripts.

How to read it:

Model	Trained on	Runs	Emergency recall	Over-triage	Macro-F1	Script gap	Recall EN / DE / HI
G1 QLoRA	all	3	0.913 ± 0.035	0.067 ± 0.011	0.812 ± 0.037	0.095 ± 0.048	0.952 / 0.929 / 0.857
G1 QLoRA	English only	1	0.595	0.032	0.717	0.452	0.857 / 0.524 / 0.405
G1 QLoRA	Hinglish only	1	0.833	0.079	0.795	0.048	0.857 / 0.833 / 0.809
B1 MuRIL	all	3	0.884 ± 0.119	0.289 ± 0.307	0.630 ± 0.120	0.040 ± 0.036	0.897 / 0.873 / 0.881
B1 MuRIL	English only	3	0.934 ± 0.057	0.406 ± 0.242	0.560 ± 0.091	0.071 ± 0.071	0.905 / 0.921 / 0.976
B1 MuRIL	Devanagari only	3	0.860 ± 0.129	0.275 ± 0.284	0.590 ± 0.071	0.190 ± 0.165	0.746 / 0.936 / 0.897
B1 MuRIL	Hinglish only	3	0.815 ± 0.095	0.198 ± 0.086	0.575 ± 0.038	0.167 ± 0.063	0.738 / 0.802 / 0.905

Table 3: Test split; the last column is emergency recall in English, Devanagari and Hinglish.

Figure file not found:
script_recall_test.png

Figure 2: Emergency recall by script for each system and training setting (test split; error bars are sd over seeds where there is more than one).

- For G1, English-only training gives Devanagari and Hinglish recall of 0.52 and 0.41, a script gap of 0.45. Hinglish-only training reaches 0.86 / 0.83 / 0.81, so romanised text appears to be a better single training script than English, though each single-script G1 is one seed and has no spread estimate yet.
- For MuRIL the seed spread (sd of 0.06 to 0.13 in recall, up to 0.31 in over-triage) is larger than most differences between training scripts, so no ranking of scripts is supported.
- All-script models see three times as many examples and update steps as single-script ones, so these runs measure script coverage and data quantity together; a matched-size control (300 examples sampled across all three scripts) has not been run.

Prompting ablation (V0). Adding 8 examples to the prompt raises emergency recall from 0.738 to 0.802 and macro-F1 from 0.547 to 0.691, and cuts the script gap from 0.333 to 0.143. Recall by script (English / Devanagari / Hinglish) is 0.905 / 0.738 / 0.571 zero-shot and 0.857 / 0.833 / 0.714 with 8 examples, so the untuned model shows the same English-over-romanised gap that motivated this project.

Not yet run. The central 2×2 ablation of safety gate and action guard (arms A0, A1, +Gate, +Guard, Full) needs the clinic simulator and guard, which are not built.

8 Challenges faced

The main difficulties so far were measurement problems, not model problems: a metric that can be gamed, unstable and non-repeatable training, and an evaluation set small enough that intervals are wide.

1. **Perfect recall from a useless model.** Our first MuRIL configuration (5 epochs, learning rate 3×10^{-5}) labelled most messages EMERGENCY. Dev emergency recall was about 1.0, but over-triage was 0.60 to 1.00 and macro-F1 0.14 to 0.54; two of the three Hinglish-only runs scored exactly like a constant EMERGENCY predictor. It was caught only because over-triage and macro-F1 are reported next to recall. Impact: those runs were discarded and the training budget increased.
2. **Unstable training.** With the final MuRIL configuration, one of three all-script seeds barely trained (final loss 0.51 against 0.02). Reporting the best seed would have hidden this.

3. **Training is not repeatable even with a fixed seed.** The same G1 configuration and seed gave macro-F1 0.810 on one run and 0.850 on another. Impact: differences of a few points between single runs are within noise.
4. **Small, clustered evaluation set.** 42 emergency cases per script on test and 18 on dev; the three renderings of a case are not independent, so intervals are bootstrapped over cases and remain wide (about ± 0.05 for recall). Dev is too small for fine threshold selection.
5. **Dataset construction.** The evaluation set was written by one person in all three forms; the first parity pass found 7 of 150 rows with unlisted details. The training set was written by a different person but within the same project, so shared style cannot be ruled out, and it contains typos and missing spaces. No clinician has reviewed the rubric or labels.
6. **Compute and tooling.** Kaggle notebooks do not allow background shell jobs and sessions can be lost, so jobs run through a Python subprocess on a chosen GPU and outputs are zipped and downloaded after each stage. The 4-bit Qwen model needs a GPU even for evaluation.
7. **Schedule.** The agent lane (simulator, tools, action guard, 2×2 arms) is behind the triage lane.

9 Mitigation plans and remaining work

The plan is to finish the triage analysis cheaply, then spend the remaining time on the agent lane, keeping the proposal's Tier 1 scope and its fallbacks.

Challenge	Mitigation	Status
Recall can be gamed	Always report over-triage and macro-F1; compare gates at a matched over-triage budget	In place
Seed instability and non-repeatable training	Three seeds, mean ± sd, no best-seed reporting; archive superseded runs	In place; single-seed ablations need more seeds
Small evaluation set	Bootstrap over cases; report intervals; keep dev for tuning only	In place; extend the test set if time allows
Writer style overlap, noisy training text	Different writers for train and test; overlap check; add programmatic romanisation noise to training text	Overlap check done; noise augmentation planned
No clinician review	Ask the TA or a qualified student reviewer to check the rubric and the emergency templates; otherwise claim agreement with a guideline-derived rubric only	Not started
Untuned probabilities	Calibrate the gate on dev and stop training on emergency recall at a fixed over-triage budget, as the proposal specifies	Planned
Agent lane behind	Build the simulator, tools and guard now; if the full scenario set does not fit, run all five arms on a smaller scenario set, as the proposal's fallback allows	Not started

Remaining work for the final submission.

1. Clinic simulator D4: synthetic patients, clinicians and appointments with a database-state success check, about 60 to 70 scenarios including identity and proxy callers, emergency interruption, medical-advice requests and adversarial cases (prompt injection, impersonation, skipping confirmation).
2. Four core tools and the deterministic action guard (allow-list, identity, confirmation, schema).
3. Orchestrator and the 2×2 system arms: A0 (plain LLM), A1 (prompted gate, no guard), +Gate (G1, no guard), +Guard (prompted gate with guard), Full (G1 with guard), all on the same backbone and token budget; unapproved-action rate as the primary metric for H2.
4. Cross-script analysis for H3 on the agent: cross-script recall gap and minimal-pair flip accuracy for the gate (computed, to be reported), plus a matched-size training control.
5. Calibration and early stopping, more seeds for the single-seed ablations, and an error analysis of 6 to 8 representative failures of G1, including difficult ones.
6. Clinician or TA review of the rubric, labels and fixed emergency templates.

Limitations to state in the final report. All data are synthetic and written by the team; the rubric is not clinician-reviewed; the test set has 42 emergency cases per script; G1 hyperparameters were not tuned while the MuRIL baseline had one round of tuning; the H1 budget was chosen after seeing results; and this is a research prototype, not a medical device.